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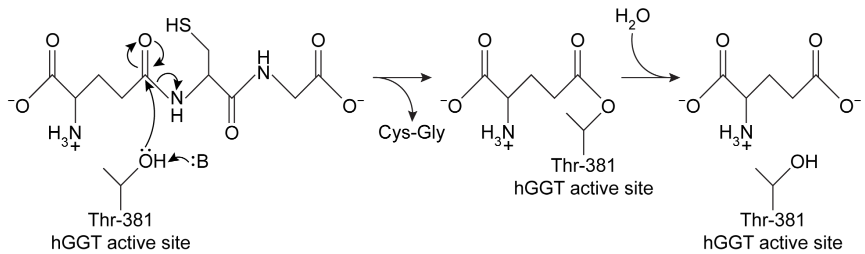


Figure 1 Proposed mechanism of hGGT-mediated GSH cleavage

Researchers interested in investigating hGGT inhibitors as a treatment for pathologies related to excess hGGT activity performed the following experiments, each containing the same concentration of hGGT.

Experiment 1

hGGT was incubated with various concentrations of GSH in the presence or absence of 250 μM candidate compound acivicin or Compound **1**. However, both candidates were found to be unsuitable as acivicin showed significant neurotoxicity and Compound **1** acted as an activator instead of an inhibitor.

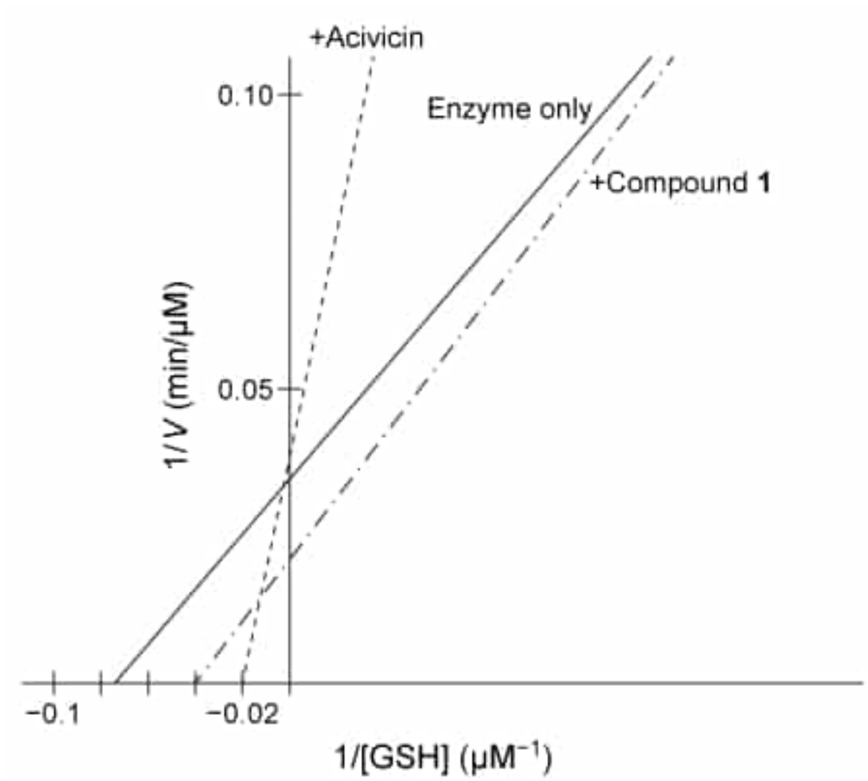


Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM

candidate inhibitors

Experiment 2

Researchers hypothesized that slight variations in the structure of Compound **1** could result in hGGT inhibition and tested several derivatives, shown in Table 1.

Table 1 Kinetic Parameters of hGGT in the Presence or Absence of Several Candidate Inhibitors at 250 μM

Inhibitor	K_M(μM)	V_{max}(μM/min)
None	13	26
Compound 2	21	21
Compound 3	6	12
Compound 4	13	14

Compound 5	9	15

Experiment 3

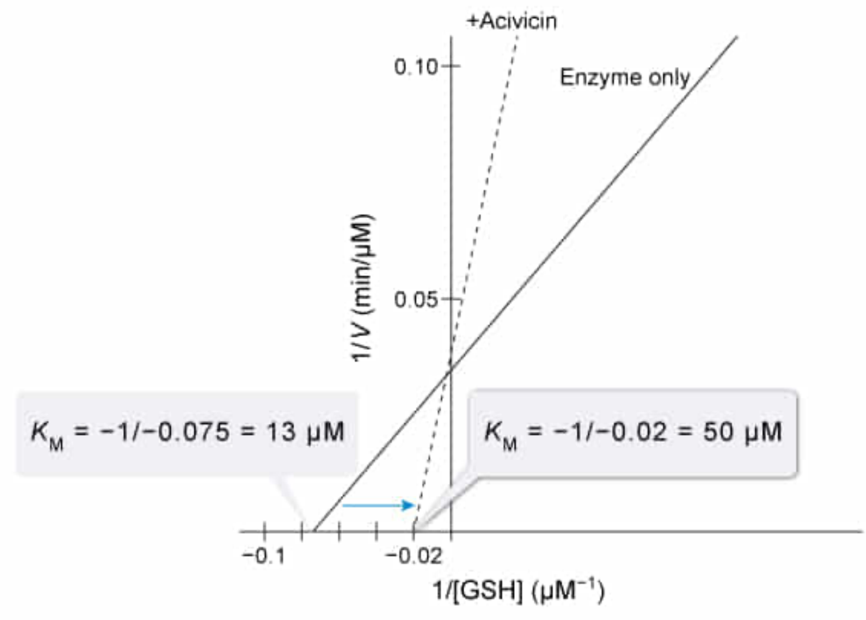
To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

Based on Figure 2, 250 μM acivicin had what effect on the K_M value of hGGT?

- A. K_M decreased to 5 μM [10%]
- B. K_M stayed constant at 13 μM [3%]
- C. K_M increased to 25 μM [20%]
- ✔ D. K_M increased to 50 μM [66%]

Correct
65% answered correctly

Explanation:



A Lineweaver-Burk plot is a double reciprocal plot that displays the **inverse of a reaction rate** (min/μM) as a function of **inverse substrate concentration** (μM^{-1}). The **Lineweaver-Burk** equation, derived from the Michaelis-Menten equation, displays the data in a linear form and facilitates the assessment of an enzyme's kinetic parameters. In this form, $\boldsymbol{V_{max}}$ is the reciprocal of the ***y*-intercept** (ie, y-intercept = $1/V_{max}$); $\boldsymbol{K_M}$ is the *negative* reciprocal of the ***x*-intercept** (ie, *x*-intercept = $-1/K_M$); and the $\boldsymbol{K_M/V_{max}}$ **ratio**, which is inversely proportional to catalytic efficiency, is the **slope** of the line.

Figure 2 shows that in the presence of 250 μM acivicin, the *x*-intercept occurs at −0.02 μM^{-1}. Therefore, the apparent K_M of hGGT under these conditions is:

$$K_M = -\frac{1}{-0.02} \mu M^{-1} = 50\ \mu M$$

In the absence of acivicin, the x-intercept occurs at approximately −0.075 μM^{-1}, giving a K_M value of about 13 μM (−1/−0.075 ≈ 13, confirmed by Table 1). Therefore, the K_M value of hGGT increased to 50 μM in the presence of 250 μM acivicin.

(Choice A) A decrease in K_M would be reflected as a left shift in the x-intercept on a Lineweaver-Burk plot. Figure 1 shows a right shift when 250 μM acivicin is added.

(Choice B) If the K_M value were unchanged, the x-intercept of the Lineweaver-Burk plot would be the same in the presence or absence of acivicin. The y-intercept was unchanged, but this indicates a constant V_{max} rather than a constant K_M.

(Choice C) A value of 25 μM is the K_M value of hGGT in the presence of 250 μM Compound **1**, not 250 μM acivicin.

Educational objective:
Lineweaver-Burk plots display the inverse of an enzymatic reaction rate as a function of inverse substrate concentration. They can be used to determine enzyme kinetic parameters with V_{max} equal to the inverse of the y-intercept, K_M equal to the negative inverse of the x-intercept, and catalytic efficiency (k_{cat}/K_M) inversely proportional to the slope of the line (K_M/V_{max}).

Subject:
Biochemistry

System:
Enzymes

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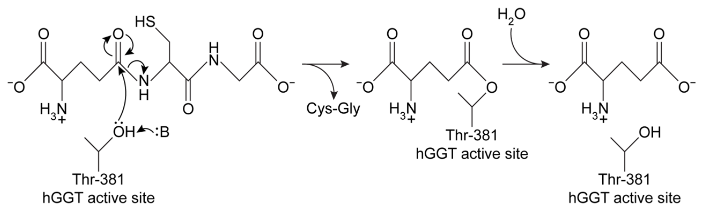

Figure 1 Proposed mechanism of hGGT-mediated GSH cleavage

Researchers interested in investigating hGGT inhibitors as a treatment for pathologies related to excess hGGT activity performed the following experiments, each containing the same concentration of hGGT.

Experiment 1

hGGT was incubated with various concentrations of GSH in the presence or absence of 250 μM candidate compound acivicin or Compound **1**. However, both candidates were found to be unsuitable as acivicin showed significant neurotoxicity and Compound **1** acted as an activator instead of an inhibitor.

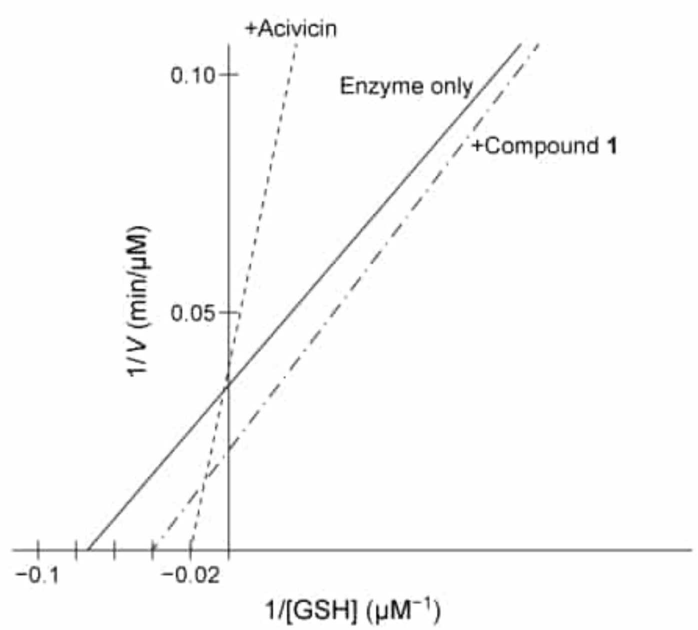

Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM candidate inhibitors

Experiment 2

Researchers hypothesized that slight variations in the structure of Compound **1** could result in hGGT inhibition and tested several derivatives, shown in Table 1.

Table 1 Kinetic Parameters of hGGT in the Presence or Absence of Several Candidate Inhibitors at 250 μM

Inhibitor	K_M(μM)	V_{max}(μM/min)
None	13	26
Compound 2	21	21
Compound 3	6	12
Compound 4	13	14
Compound 5	9	15

Experiment 3

To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

Which compound in Table 1 acts as an uncompetitive inhibitor?

- A. Compound **2** [12%]
- ✔ B. Compound **3** [64%]
- C. Compound **4** [16%]
- D. Compound **5** [5%]

Correct
65% answered correctly

Explanation:

Explanation:

Inhibitor	K_M (μM)	V_{max} (μM/min)	V_{max}/K_M(min^{-1})
None	13	26	2
Compound 2	21	21	1
Compound 3	6	12	2
Compound 4	13	14	1.1
Compound 5	9	15	1.7

Enzyme inhibitors can act competitively, uncompetitively, or as a hybrid of the two (ie, mixed). Each class of inhibitor has its own set of unique effects on enzyme kinetics.

Uncompetitive inhibitors can bind to enzymes only after the enzyme has bound its substrate. The enzyme is then unable to convert substrate to product, resulting in a **decrease in V_{max}**. As the inhibitor binds the enzyme-substrate (ES) complex, it effectively depletes ES levels by

forming enzyme-substrate-inhibitor (ESI) complexes. This shifts the equilibrium such that additional free enzyme is induced to bind substrate, causing an apparent **decrease in K_M**. In uncompetitive inhibition, K_M and V_{max} both decrease by **exactly the same factor**, and therefore the **V_{max}/K_M ratio is unchanged**, as reflected by the parallel slopes in Lineweaver-Burk plots.

Table 1 shows that in the absence of any inhibitor, hGGT has a K_M of 13 μM and a V_{max} of 26 μM/min, giving a V_{max}/K_M ratio of 2 min^{-1}. The only inhibitor in the table that yields the same ratio is Compound **3**, with a K_M of 6 and a V_{max} of 12. Therefore, **Compound 3 acts as an uncompetitive inhibitor**.

(Choice A) Compound **2** increases the K_M value of the enzyme while decreasing V_{max}, making it a mixed inhibitor with a higher affinity for free enzyme than for the ES complex.

(Choice C) Compound **4** does not alter the K_M value of hGGT, and therefore it acts as a special class of mixed inhibitor called a noncompetitive inhibitor. It is *not* an uncompetitive inhibitor.

(Choice D) Although Compound **5** decreases both V_{max} and K_M, it is *not* an uncompetitive inhibitor because it decreases V_{max} by a *greater factor* than K_M, yielding an *altered* V_{max}/K_M ratio. Uncompetitive inhibitors *always* decrease V_{max} and K_M by *exactly* the same factor, yielding *no change* in the V_{max}/K_M ratio. Compound 5 is a mixed inhibitor with a higher affinity for ES complex than for free enzyme.

Educational objective:
Different enzyme inhibitors can act competitively, uncompetitively, or as a mix of the two. Uncompetitive inhibitors bind only the enzyme-substrate complex and decrease both the V_{max} and K_M values of an enzymatic reaction by the same factor, resulting in an unchanged V_{max}/K_M ratio.

Time spent:0 sec
Subject:
Biochemistry

QID:401560
System:
Enzymes

Human gamma-glutamyl transpeptidase (hGGT) is an enzyme involved in cysteine homeostasis, and its overexpression has been linked to asthma, cancer, and other diseases. It catalyzes the addition of water across a bond to cleave extracellular glutathione (GSH) into glutamate and a dipeptide composed of cysteine and glycine. It is believed to function by the mechanism shown in Figure 1.

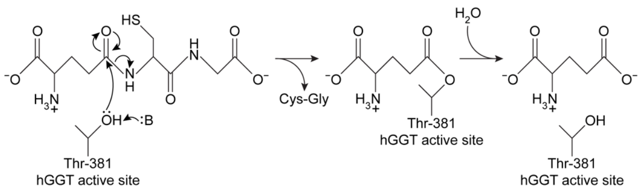

Figure 1 Proposed mechanism of hGGT-mediated GSH cleavage

Researchers interested in investigating hGGT inhibitors as a treatment for pathologies related to excess hGGT activity performed the following experiments, each containing the same concentration of hGGT.

Experiment 1

hGGT was incubated with various concentrations of GSH in the presence or absence of 250 μM candidate compound acivicin or Compound **1**. However, both candidates were found to be unsuitable as acivicin showed significant neurotoxicity and Compound **1** acted as an activator instead of an inhibitor.

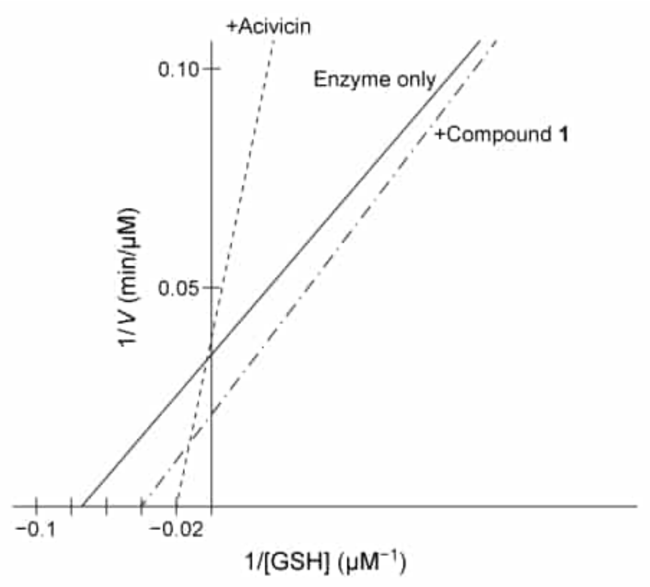

Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM candidate inhibitors

Experiment 2

Researchers hypothesized that slight variations in the structure of Compound **1** could result in hGGT inhibition and tested several derivatives, shown in Table 1.

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Inhibitor	K_M(μM)	V_{max}(μM/min)
None	13	26
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Experiment 3

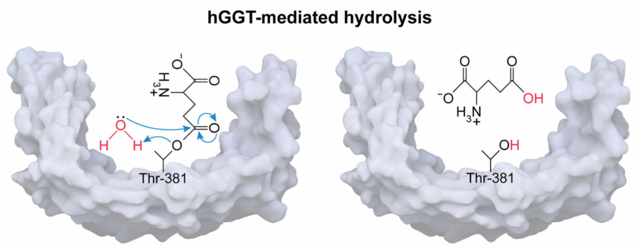
To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

In the reaction observed in Figure 1, hGGT acts as:

- A. a hydrolase. [75%]
- B. a transferase. [5%]
- C. an isomerase. [1%]
- D. a lyase. [17%]

Correct
76% answered correctly

Explanation:



Enzymes are broadly divided into six classes based on the **type of reaction** they catalyze. Several classes of enzymes can facilitate the formation or breakage of chemical bonds, but different classes of enzymes use different methods to do so. Hydrolases **break bonds** by **adding water** across them in a **hydrolysis** reaction.

In Figure 1, hGGT is shown to cleave GSH into Cys-Gly and glutamate, and adds water in the final step. The net effect of the reaction is cleavage of a peptide bond by the **addition of water**. Therefore, hGGT **acts as a hydrolase** in the reaction shown in Figure 1.

(Choice B) Transferases catalyze the transfer of a chemical group from one molecule to another. To act as a transferase, hGGT would have to transfer Cys-Gly or glutamate to another molecule rather than just release them. However, this transfer is not shown in the reaction given in Figure 1.

(Choice C) Isomerases rearrange the chemical groups within a molecule rather than divide one molecule into two.

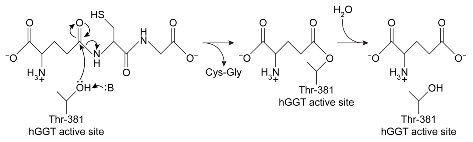
(Choice D) Lyases break bonds in a molecule but do so *without* the addition of water. Lyase activity typically results in the formation or breakage of a double bond. Because water was added in the given reaction, the enzyme hGGT cannot be a lyase.

Educational objective:
Enzymes are broadly grouped into six classes: oxidoreductase, transferase, hydrolase, lyase, isomerase, and ligase. Hydrolases break bonds by adding water across them.

Time spent:0 sec
Subject:
Biochemistry

QID:401561
System:
Enzymes

Human gamma-glutamyl transpeptidase (hGGT) is an enzyme involved in cysteine homeostasis, and its overexpression has been linked to asthma, cancer, and other diseases. It catalyzes the addition of water across a bond to cleave extracellular glutathione (GSH) into glutamate and a dipeptide composed of cysteine and glycine. It is believed to function by the mechanism shown in Figure 1.



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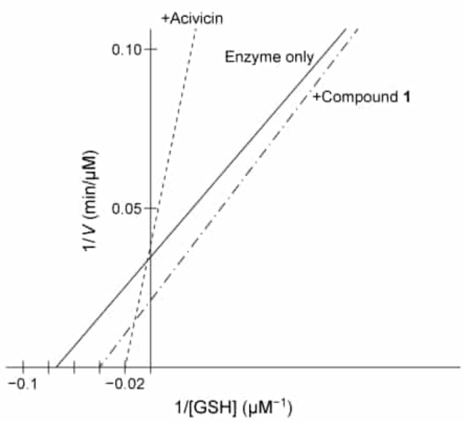

Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM candidate inhibitors

Experiment 2

Researchers hypothesized that slight variations in the structure of Compound **1** could result in hGGT inhibition and tested several derivatives, shown in Table 1.

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Experiment 3

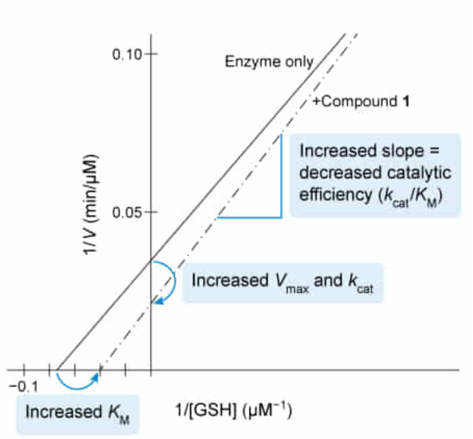
To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

Based on Figure 2, Compound **1** activates hGGT in which of the following ways?

- A. It increases the turnover number. [48%]
- B. It decreases the Michaelis constant. [10%]
- C. It increases the k_{cat}/K_M ratio. [28%]
- D. It decreases the maximum velocity. [12%]

Correct
48% answered correctly

Explanation:



Activators can enhance enzyme activity by increasing the turnover number k_{cat} to achieve a higher maximum rate (V_{max}), by decreasing the Michaelis constant K_M to reach the maximum rate at lower substrate concentrations, or both (increased catalytic efficiency, k_{cat}/K_M). A decrease in k_{cat} can be offset by a greater decrease in K_M, and an increase in K_M can be offset by a greater increase in k_{cat}.

Lineweaver-Burk plots display the relationship between the **inverse of a reaction rate** and the **inverse of the initial substrate concentration** for enzymatic reactions. This relationship is linear, and V_{max} and K_M can be determined from the *y*-intercept and *x*-intercept, respectively. Catalytic efficiency and k_{cat} can also be determined, as $k_{cat} = V_{max}/[\text{enzyme}]_{tot}$, and catalytic efficiency is the ratio k_{cat}/K_M.

However, because Lineweaver-Burk plots relate the *inverse* of these parameters, the **trends shown in the plot** represent the **inverse** of the trends of the enzyme-substrate system. For example, a decrease in the *y*-intercept corresponds to an *increase* in V_{max}, and vice versa.

Figure 2 shows that in the presence of Compound **1**, the *y*-intercept decreases; therefore, Compound **1** causes an *increase* in V_{max} **(Choice D)**. The passage states that all experiments were carried out at the same enzyme concentration, and therefore V_{max} directly corresponds to the turnover number k_{cat}. Because $\boldsymbol{V_{max}}$ **increases** in the presence of Compound **1**, so does the **turnover number**.

(Choice B) Figure 2 shows a decrease in the magnitude of the *x*-intercept in the presence of Compound **1**, which corresponds to an *increase* in the Michaelis constant K_M.

(Choice C) Figure 2 shows an increase in the slope of the line in the presence of Compound **1**, indicating an increased K_M/V_{max} ratio. This is the same as a *decreased* V_{max}/K_M ratio, which is directly proportional to the catalytic efficiency, k_{cat}/K_M. Therefore, Compound **1** causes a *decreased* k_{cat}/K_M ratio.

Educational objective:

Lineweaver-Burk plots display the inverse of reaction velocity as a function of inverse substrate concentration. As a result, the trends in Michaelis-Menten parameters are inverted on the graph. An increase in *y*-intercept corresponds to a decrease in V_{max} and, if enzyme concentration is constant in all experiments, a decrease in the turnover number k_{cat}.

Time spent: 0 sec	**QID**:401562
Subject: Biochemistry	**System:** Enzymes

Human gamma-glutamyl transpeptidase (hGGT) is an enzyme involved in cysteine homeostasis, and its overexpression has been linked to asthma, cancer, and other diseases. It catalyzes the addition of water across a bond to cleave extracellular glutathione (GSH) into glutamate and a dipeptide composed of cysteine and glycine. It is believed to function by the mechanism shown in Figure 1.

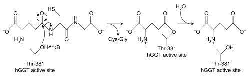


Figure 1 Proposed mechanism of hGGT-mediated GSH cleavage

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Experiment 1

hGGT was incubated with various concentrations of GSH in the presence or absence of 250 μM candidate compound acivicin or Compound **1**. However, both candidates were found to be unsuitable as acivicin showed significant neurotoxicity and Compound **1** acted as an activator instead of an inhibitor.

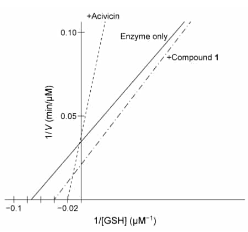


Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM candidate inhibitors

Experiment 2

Researchers hypothesized that slight variations in the structure of Compound **1** could result in hGGT inhibition and tested several derivatives, shown in Table 1.

Table 1 Kinetic Parameters of hGGT in the Presence or Absence of Several Candidate Inhibitors at 250 μM

Inhibitor	K_m(μM)	V_{max}(μM/min)
None	13	26
Compound 2	21	21
Compound 3	6	12
Compound 4	13	14
Compound 5	9	15

Experiment 3

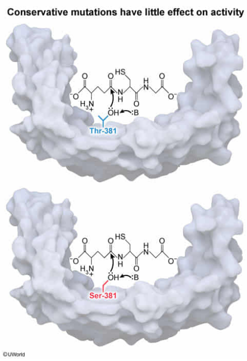
To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

Mutating residue 381 to which of the following amino acids would most likely result in the smallest change in hGGT activity?

- A. Glycine [4%]
- B. Lysine [3%]
- ✔ C. Serine [88%]
- D. Cysteine [3%]

Correct
88% answered correctly

Explanation:



Mutating an amino acid in an enzyme can have an **effect on the enzyme's activity** if the function performed by that amino acid is abolished. Conservative mutations replace one amino acid with another that has **similar properties** and are less likely to have an effect on activity. The more similar two amino acids are (ie, more conservative), the less likely that replacing one with the other will alter protein function.

Figure 1 shows that the hGGT-mediated reaction relies on a threonine residue at position 381 using its hydroxyl group to nucleophilically attack GSH at a carbonyl. **Serine and threonine are nearly identical** and differ only in that threonine has a methyl group where serine has a hydrogen atom. Both have a hydroxyl group approximately the same distance away from the α-carbon of the amino acid backbone. Therefore, serine is likely to participate in a similar nucleophilic attack on GSH. Although a mutation from threonine to serine at position 381 might alter enzyme activity, it is the least likely of the available choices to result in a significant change.

(Choice A) Glycine has a hydrogen atom for a side chain and would not be able to attack a carbonyl on GSH.

(Choice B) Lysine is much larger than threonine and has an amino group rather than a hydroxyl group on its side chain. Its substitution at position 381 would likely alter enzyme activity significantly.

(Choice D) Cysteine is similar to serine but has a thiol group instead of a hydroxyl group. Although cysteine can facilitate proteolysis, its substitution at position 381 would most likely cause a significant difference in activity.

Educational objective:
Mutations in enzymes can cause a change in enzymatic activity. Conservative mutations (substitution of one amino acid with another that has similar properties) are less likely to result in changes in activity.

Time spent:0 sec
Subject:
Biochemistry

QID:401563
System:
Enzymes

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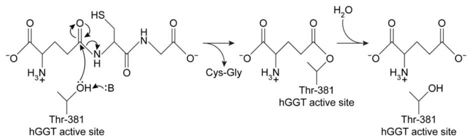

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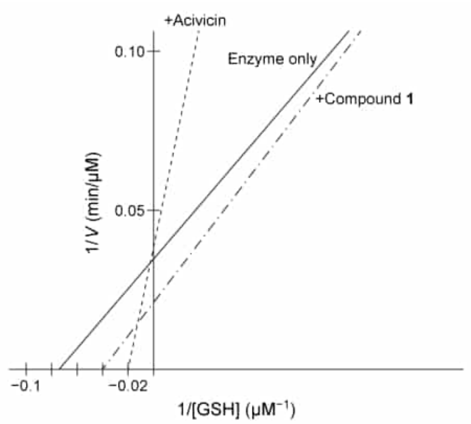

Figure 2 Lineweaver-Burk plot of hGGT activity with respect to GSH in the presence or absence of 250 μM candidate inhibitors

Experiment 2

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Compound 5	9	15

Experiment 3

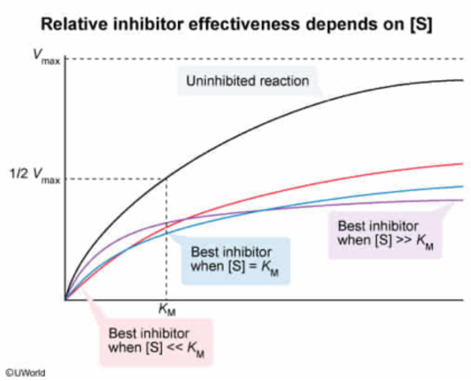
To better understand the physiological conditions under which hGGT operates, researchers exposed a cultured cell line expressing hGGT on the cell surface to a solution that mimicked the composition of extracellular fluid, including levels of GSH. They found that in this setup, uninhibited hGGT results in a reaction rate that is approximately ½V_{max}.

Based on the information in Experiment 3, if researchers wish to further study the inhibitor that is most effective under physiological conditions, they should choose the one that yields the lowest hGGT activity when GSH concentration is:

- A. orders of magnitude less than the K_M of the uninhibited enzyme. [16%]
- ✔ B. approximately equal to the K_M of the uninhibited enzyme. [48%]
- C. approximately four times the K_M of the uninhibited enzyme. [10%]
- D. orders of magnitude greater than the K_M of the uninhibited enzyme. [24%]

Correct
47% answered correctly

Explanation:



The rate of an enzymatically catalyzed reaction is **dependent on substrate concentration**, which can be **expressed in terms of multiples of K_M**. The reaction velocity for any enzymatic reaction can be expressed as a **percentage of V_{max}** such that if substrate concentration is equal to nK_M, the reaction runs at $n/(n+1)$ V_{max}. For example, when substrate concentration is equal to K_M, the reaction rate is 1/(1+1) V_{max} = ½V_{max}.

When substrate concentration is equal to 2K_M, the reaction rate is 2/(2+1) V_{max} = (2/3)V_{max}, and so on. Varying the substrate concentrations alters the reaction velocity by changing the concentration of the enzyme-substrate complex.

The passage states that in a setup that mimics physiological conditions, the reaction **runs at approximately ½V_{max}**, and therefore the **physiological concentration of GSH** under these conditions must be **approximately equal to the K_M** of the uninhibited enzyme. Some inhibitors are more effective at certain substrate concentrations than others, and an inhibitor that works well under saturating conditions might not work as well when substrate concentration is near K_M. Therefore, if researchers wish to use the inhibitor that is most effective at physiological conditions, they must use the one that works best at the **physiological substrate concentration**. In this case, they should determine which inhibitor results in the lowest activity when substrate concentration equals the K_M of uninhibited hGGT.

(Choice A) A substrate concentration that is orders of magnitude lower than K_M would be significantly lower than physiological concentration, so an inhibitor that works well at this concentration might not work well under physiological conditions.

(Choices C and D) A substrate concentration that is four or more times greater than K_M is substantially higher than physiological concentration, so an inhibitor that works well under these conditions might not work well in a physiological setting.

Educational objective:

Enzymatic reaction rate can be expressed as a fraction of V_{max} and is directly related to substrate concentration, which can be expressed as a multiple of K_M. Inhibitors that work well at low substrate concentrations ([S] << K_M) might not work as well at higher concentrations, and vice versa.

Time spent: 0 sec	**QID**:401564
Subject: Biochemistry	**System:** Enzymes